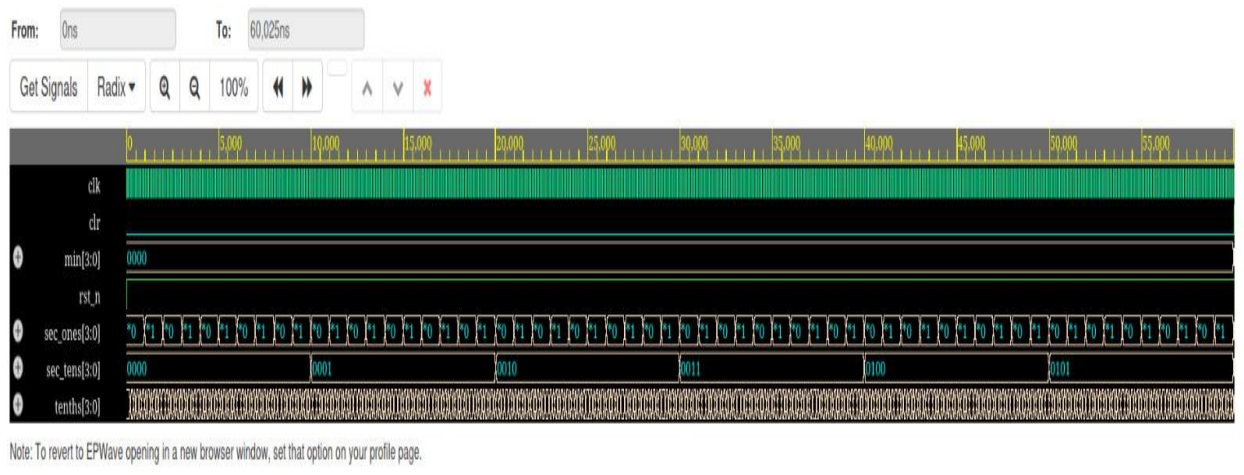


Task#13 Stopwatch

EDA PLAYGROUND LINK

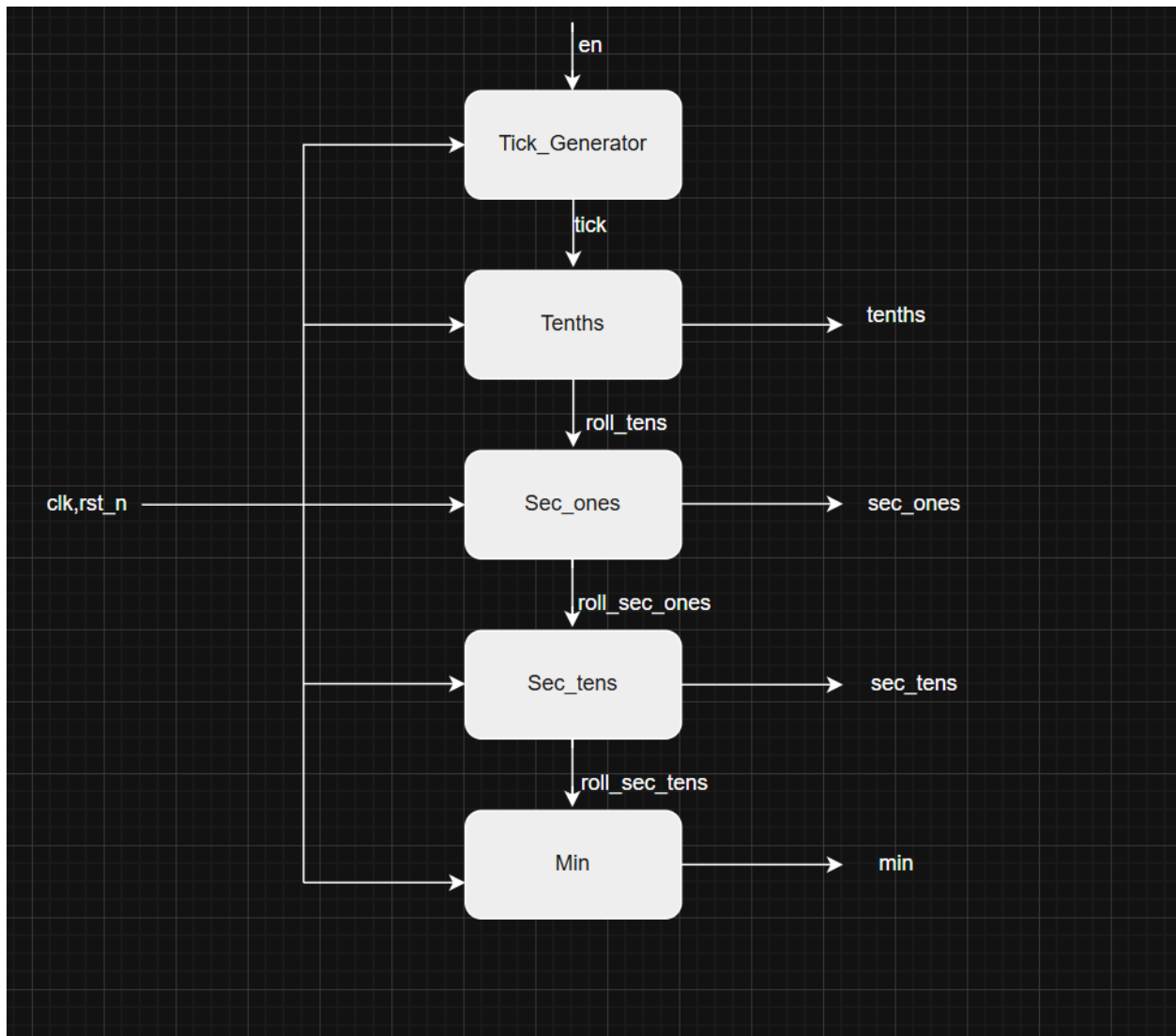
<https://edaplayground.com/x/DWyy>

SCREENSHOTS



```
Chronologic VCS simulator copyright 1991-2025
Contains Synopsys proprietary information.
Compiler version X-2025.06-SP1_Full64; Runtime version X-2025.06-SP1_Full64; Jul 23 05:41 2026
[PASS] starts at 0:00:0
after 1 minute: minute=1 sec_tens=0 sec_ones=0 tenths=0
[PASS] minute rollover correct
[PASS] clr resets all digits
$finish called from file "testbench.sv", line 40.
$finish at simulation time          60025
      V C S   S i m u l a t i o n   R e p o r t
Time: 60025 ns
CPU Time:      0.490 seconds;      Data structure size:  0.0Mb
Thu Jul 23 05:41:30 2026
Finding VCD file...
```

BLOCK DIAGRAM



CALCULATION

Simulation clock: 100 MHz, period = 10 ns (always #5 clk = ~clk;)

Required tick frequency: 1 tick every 0.1 s $\rightarrow f_{\text{out}} = 1 / 0.1 = 10$ Hz

Divisor needed:

$$N = f_{\text{in}} / f_{\text{out}} = 100,000,000 \text{ Hz} / 10 \text{ Hz} = 10,000,000$$

So the internal counter must count from 0 to 9,999,999 (10,000,000 states) before pulsing — requiring $\lceil \log_2(10,000,000) \rceil = 24$ bits of counter width.

Duty cycle of the tick pulse: the pulse is high for exactly 1 clock cycle out of every 10,000,000, so:

Duty cycle = $1 / N = 1 / 10,000,000 = 0.00001\%$ (essentially a single-cycle strobe, not a square wave)

Pulse width:

Pulse width = 1 clock period = 10 ns